

Occupational safety

Define as the right way of doing things.

General workshop safety

It's essential to be aware of and able to foresee the dangers which exist in the workshop and are likely to affect your health and safety.

You should make sure that your workspace is as safe as possible so that few dangers arise.

- i) Use the appropriate protective clothing and equipment to minimize the risk of accident
- ii) Act in a safe manner at all times.

Causes of accidents

1. Causes due to human beings

a) Carelessness

This is due to overconfidence, loss of interest, fatigue (continuous work without rest), monotony, unnecessary emotion which diverts concentration on the work being done hence resulting to accidents.

b) Ignorance

An operator must understand the users and the function of his machine equipment.

c) Unsuitable clothing in working areas,

Personnel should wear clothing which will provide protection against dangers in areas where they work.

Shoes. They should be a type which provides protection against slippery conditions.

They should be strong enough in the feet to prevent injury from hurting objects e.g. safety boots.

Gloves- They should be worn where there are risks when handling sharp objects whether inside or outside buildings.

Safety glass

- Safety glasses or goggles should be worn where there's likely of danger to the eyes, such as using drills, grinding machines etc.

d) Untidiness

- keep all passage ways clear and clean.

Circular rods if stepped upon can cause nasty falls.

-keep the workshop floor free from grease and oils.

2. Contributing causes

a) Unsatisfactory hand tools

E.g. being worn out without handles

-hammers with loose handles or crippled faces should never be used.

-Chisels with burred heads are dangerous as broken pieces from the edges may fly off and cause injury.

-files should have tight handles to prevent accidents to the hands.

-spanners which fit incorrectly will slip and can result in damaging fingers.

-when drilling always ensure that the work is securely held in a vice or otherwise securely clamped.

b) Unsatisfactory machines

- That is being insufficiently spaced, unguarded, broken, improperly adjusted, insufficiently lubricated

And being too small or weak for the job being undertaken.

-so see that all machines are properly guarded, all abrasive wheels are kept in good condition and run at safe speeds.

- don' t attempt to adjust or remove by hand a belt on a rotating shaft.

-don' t allow idle belt to rest on rotating shaft and don' t attempt to oil overhead shafting while it' s in motion.

c) Physical condition of personnel while working

E.g. before lifting equipment ensure that the weight is within your capability to handle. A base that any thing used for lifting is serviceable and in good working condition chains, hooks etc

Should never be overloaded.

Industrial safety

The factory act 1961 states that:

- i) Floors steps stairs passages and gang ways must be soundly constructed properly maintained and kept free from obstruction and any substance likely to cause a person to slip.
- ii) Hand rails must be provided for stairs.
- iii) All ladders must be soundly constructed and properly maintained.
- iv) Opening in floors shall, wherever practicable be securely fenced.

Special regulations

The following precautions should be observed

- i) Always work never runs
- ii) Never throw rubbish on the floor
- iii) Keep gang ways and work areas free of metal bars, components, etc.
- v) Keep to gangways when moving about never takes short cuts.
- vi) If oil water or grease is spilled wipe it up immediately
- vii) Wear safety shoes- shoes are available with anti slip soles and with metal toe caps to protect the feet from falling objects.
- viii) Always check ladders for damage before use.

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- ix) Always position ladders on firm base at the correct angle.

Factories act 1948 regarding safety

The factories act was passed by the governor general of India on 23rd sept. 1948. The act was further amended in 1950, 1951, 1954, & 1976 which came into force on 26th Nov. 1976

Aims and objectives

The major objectives of the factory act are;

1. To provide protection to the workers employed in factories against industrial hazards and to ensure safe and better working conditions.
2. It regulates and maintains properly various safety health and welfare activities in the factories.
3. It regulates & maintains working hours of workers, employment of children and adolescents, employment of women, annual leave with wages etc.

The salient features of this act regarding safety are as follows:

1. Fencing of machinery

-Every prime mover like engine or motor, moving part of the machinery is fenced properly.

2. Work on or near the machinery in motion.

Only specially trained workers wearing tight fitting clothes should carry out inspection of any part of the machinery in motion.

3. Employment of young persons on dangerous machines.

No young person should be allowed to work on dangerous machine unless he is properly trained and carefully supervised

4. Hoist and lifts

-Every hoist and lifts should be of good mechanical construction, adequate strength and must be protected by enclosure and fitted with gates.

- Every hoist and lift should be adequately maintained and periodically examined

5. Lifting machinery, chains, ropes and lifting tackles

Lifting machines such as cranes, crab, etc should be of good construction, adequate strength should be maintained.

6. Revolving machinery

It should always be ensured that safe working peripheral speed is not exceeded for every revolving machinery.

7. Pressure plants

- The pressure plant should not be operated at a pressure higher than the specified safe working pressure.

8. Floors stairs and means of access to different places

All floors, steps, stairs, passages should of sound construction and free from obstructions

9. Pits sumps, opening in floor etc.

-Every pit, sumps, opening in floor, fixed vessels, tanks etc should be securely covered or fenced.

10. Excessive weights

- No person should be asked to lift carry or move any load so heavily that' s likely to cause him injury.
- Protection of eyes.

Classification of fires

Fire can cause loss of lives, jobs equipment, materials and buildings.

For all practical purposes there are three main classes of fires; A B&C

Class A fires.

Fires in ordinary combustible materials such as wood, paper & cloth.

The majorities of fires are in this class and can be most effectively extinguished by cooling with water.

Class-B fires

- Fires from flammable liquids such as paraffin, petrol, paint, varnishes, oil and from greases and fats. These are effectively extinguished by smattering foam or other agents which exclude oxygen.

Class-C fires

-fires involving live electrical equipment and wiring. The safest method of extinguishing is to displace the oxygen by projecting a gas or dry powder into the vicinity of the fire.

In the event of a fire:

The following precautions should be observed at al times.

- i) Know the correct fire drill and the positions of fire alarms, fire fighting equipment and emergency exits.
- ii) Know the correct appliance to use for a particular type of fire and know how to use it.
- iii) Don't block fire exits.
- iv) Don't use fire appliances for any purpose other than intended.
- v) Never smoke in no smoking areas
- vi) Always ensure that matches and cigarettes are put out before throwing them away
- vii) Avoid spillage of flammable liquids.

Extinguishers and fighting procedures

.i) water type

These are usually operated by striking a plunger to release a gas which pressures the water and forces it out of the nozzle. They are stable only for class-A fires.

-the jet should be directed at the base of the fire and be kept moving across the area of the fire.

2. Chemical foam type

-these may be operated by inverting the extinguisher or by releasing a plunger allowing two solutions to mix and produce foam. They are used on class B fires involving flammable liquids.

-the jet is directed with a gentle sweeping movement, allowing the foam to drop down and lie on the surface of the liquid. This smothers the fire by excluding oxygen.

3. carbon-dioxide (CO₂) type

These are operated by squeezing the discharge lever which allows the liquid CO₂ to be discharged as a gas. They are used on class C fires where after the current has been switched off the discharge horn is directed straight at the fire.

4. Dry powder pressure type.

This type contains a pressurized gas and a dry chemical agent. After the release lever has been operated discharge is controlled by a nozzle at the end of the hose.

These are used on class C fires in the same way as the CO₂ type.

Use of fire extinguishers

Class of fire		Type of fire extinguisher			
		Water	Foam	CO ₂	Dry powder
A	ordinary fires: wood , paper ,cloth	Yes	Yes	No	No
B	Flammable liquids: petrol ,paraffin	No	Yes	Yes	Yes
C	Electrical equipment	No	No	Yes	Yes

Electrical safety

-Electrical can't be seen we only see the effect due to it.

-Because of this care must be taken when using electricity. There are two main causes of accidents due to electricity.

These are

(i) Carelessness: this arises because of laxity. When electricians have worked for long period they tend to assume the high standards of safety required and start to maintain equipment when it's live and as a result an accident can easily occur.

ii) Ignorance and inexperience.

Most of the electrical tasks appear very simple when carried out by well trained electricians. When UN qualified personnel attempt to do the same they are usually unable and they leave the tasks uncompleted. This is dangerous to the others. The safety precaution in any electrical installation is to:

- i) Prevent electric shock.
- ii) Prevent the occurrence of fire due to electrical fires.

Electric shock

-One gets electric shock when he or she is in contact with two objects that are at different potentials.

-the person complete the circuit to earth and the current flows through him/or her.

The damage done to the human body will depend on the following factors:

- i) Voltage between the two points
- ii) The amount of current flowing
- ii) The time taken for the current to flow

-the methods used to prevent electric shock are;

- i) Earthing the metallic and any equipment within the installation
- ii) Using all the insulated wiring systems

Treatment of electric shock

If somebody get electric shock, the immediate action should be:

- i) Switch off the supply
- ii) If necessary start artificial respiration
- iii) Seek medical assistance

iv) Treat the burns or injuries.

v) Keep the victim warm.

If the person is in contact with the live conductors then the first thing is to break the contact .care must be taken in order not electrocute yourself.

Try to switch off the supply or to unplug the equipment incase it' s a portable tool e.g. a drill. Under no circumstances should you attempt to touch the person. It may be possible to detach the person by pulling or pushing him/her by using insulated object e.g. an insulated cable looped round the victims arm or body using a dry wooden pole.

- After the victim is removed from the live contacts provide artificial respiration by using any of the following methods

i) Mouth to mouth

ii) Holger-Nelson

i)Mouth -mouth

In this method, the patient must be on his back and his mouth should be inspected for any obstructions e.g. false teeth. The patient head should be extended by placing one hand on his crown and the other immediately beneath his chin. Gently bend his head backwards and then using his both hands, lift his jaw forward. Place your lips over the patients' mouth and make a good seal with the thumb and one fore finger of one hand gently close the patient nostrils. Take a deep breath and exhale into the patient using little force. Watch and feel the patient' s chest rise. Remove your lips and let the lungs deflate. The above procedure should be repeated twelve times per minute remembering to turn away your head as you don' t fill your own lungs with the patients expired air. Continue inflation and depletion of the lungs until spontaneous breathing is maintained.

ii) Holger and Neilson.

The patient should be placed face downwards with the arm overhead, the elbows flexed so that one hand rests on the other in turn. Turn the patients head to one side so that the neck rests on his upper most hand. Kneel to one side of the patients head and put the foot of your other leg near his elbow. Place your hands on his back just below the shoulder blades and rock forwards with your elbow straight until your arms are approximately vertical, exerting- steady pressure on his chest.

Grasp the patient' s arms just above the elbow and rock backwards raising his arms until tension is felt at the patients shoulder lower his arms. The complete cycle should be repeated twelve times a minute. This method is practical only when there' s no gross injury to the arms shoulder and the ribs.

Workshop layout

Workshop is a place where products are manufactured.

Shop layout

- It's the proper arrangement of tools and equipments to achieve the efficient use of machines, equipments and to facilitate the processing of the product in most efficient and economical manner.

Factors

The layout of shop depends on the following factors

1. Size and types of components to be produced.
2. Types of machines available
3. Number of machines available
4. Sequence of operations
5. Power requirement
6. Store location
7. Inspection
8. Utility of individual machine
9. Interior and exterior set up of the shop

Location

The location of the shop depends on

1. Decision of a new product
2. Nearness to raw material
3. Availability of space power labour and water
4. Financial and other aids
5. Facilities for expansion

Principles of shop planning

1. Material handling

The movement of workers and materials should be minimum

-Smooth and continuous flow- means bottlenecks, congestion points and tracking should not exist in a workshop.

2. Storage.

Space utilization- the total space available should be utilized properly. Raw material and products should be stored in racks if possible.

3. *Safety*-The workplace should be free from pollution and safe. It should have proper light ventilation and other amenities.

4. *Aesthetics*

5. *Plan of workmanship*

-for convenience of the instructor during supervision of student, demonstration and working in the workshop.

-workshop should accommodate different age groups and physical abilities.

6. *Machine layout*

-equipment and machine purchased should be of high quality but not excessively expensive.

-shop equipment should be arranged to make maximum use of lighting.

7. *Electrical supply*

-shop should be sufficiently provided with light and should be well diffused.

-the lighting can be incandescent or vapour lamp and should be semi-direct or indirect light type.

-the light switches should be located near entrance exists and wired so that part of the work can be switch on and off.

-All lighting fixtures used for closed work require individual control so that they can be turn off when not in use.

MATERIALS AND PROCESSES

-Many things and items that mankind use is made of materials.

-such materials are metal, plastics, glass, rubber, wood, oil, stones etc are used in the manufacture of many kinds of things which help mankind to be comfortable.

-Different materials are used to make or manufacture different kinds of things due to their different properties.

Metals and non metals

i) Metals – are polycrystalline bodies consisting of a great number of fine crystals of metallic materials.

ii) Non – metals – are non metallic materials e.g. leather rubber asbestos and plastics.

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-metals are obtained by the process known as the metallic ore is heated in a suitable furnace and the metal is separated from all the

METALS

Metals are divided into two groups

- a) The ferrous metals – are those metals which contain a large percentage of iron. These are iron and steel.
- b) The non-ferrous metals are those which contain a small percentage of iron as an impurity or contain no iron at all e.g. copper, brass, and tin, aluminum, bronze etc.

iii) Alloys – is a metal composed of two or more metals.

-it usually consists of a base metal (the largest part of the alloy) and smaller amount of other metals.

E.g. brass is an alloy of copper (being the base metal) and zinc.

Bronze is an alloy of copper (the base metal) and tin.

Non-ferrous

Iron

Its not used in its pure state

steel

is an alloy of iron and carbon

use for making;
-sheet of metals

-hand tools
-bars of various shapes
-rods and wires etc

Ferrous

Non-ferrous

Copper

Use

-Radio parts
-Wires
-Soldering bits
-Rivets etc

Aluminum

Use

-Cooking utensils
-Electric wires
-Rivets etc

Tin

Use

-rarely used in its pure state
-used in tin plate
-used with other metals to form alloys e.g. bronze

Mechanical Properties of materials

i) Ductility

A ductile material is one which can be drawn out permanently by a tensile force.

This property is particularly important to the production engineer when he makes wire or when he forms steel sheet to produce a motor – car body.

- It is the ability of a metal to be drawn out into fine wire without breaking.

ii) Toughness

A tough material resists fracture by blows because considerable energy is required to crack or break it. Tough materials have a high tenacity combined with good or fair ductility.

Toughness decreases with heating.

iii) Strength – defined as the ability of the material to sustain loads without distortion.

iv) Hardness – it's the ability of a material to be hammered or rolled out into thin sheets without cracking.

-very few materials have good malleability when cold but most are malleable when heated to a suitable temperature.

v) Corrosion – is the gradual deterioration of material by chemical reaction with its environment.

vii) resistance- it' s the property of a material due to which it resists the flow of electricity through it.

viii) Heat treatment- is defined as an operation or combination of operations involving heating and cooling of a metal or alloy in the solid state with the purpose of changing the properties of the materials.

Heat treatment consists of 3 phases

- i) heating of the metal
- ii) soaking of the metal
- iii) cooling of the metal

Purpose of heat treatment

-to improve mechanical properties e.g. tensile strength,

Ductility, hardness, shock resistance, resistance to carbon etc.

Materials and processes

-there are many different types of metals that are used to make articles of various uses and of different shapes.

-the qualities that metals have which determine whether they can be made into various shapes or not, drawn into wires and whether brittle/hard or not are called the properties of metals.

-metals have many physical and mechanical properties which affect their working qualities.

Physical properties are;

- a) *Density* is the heaviness of a metal e.g. Aluminum has lower density than copper.
- b) *Conductivity* is the ability of metal to conduct heat and electricity e.g. copper and silver are the best conductors of both heat and electricity
- c) *Magnetism* is the property to attract iron, nickel and cobalt. Metal which has the ability to attract or repel other metals is called a magnet.
- d) *Fusibility* is the property of a metal to become liquid when heated. different metals have different melting points e.g. mild steel is between 1300 c and Aluminum is 600 c
- e) *Color*: various metals have different colors aluminum is silvery white while copper is brownish.
- f) *Structure*: this is how the small particles in a metal are arranged. Different metals have different atomic arrangements.

Extraction process

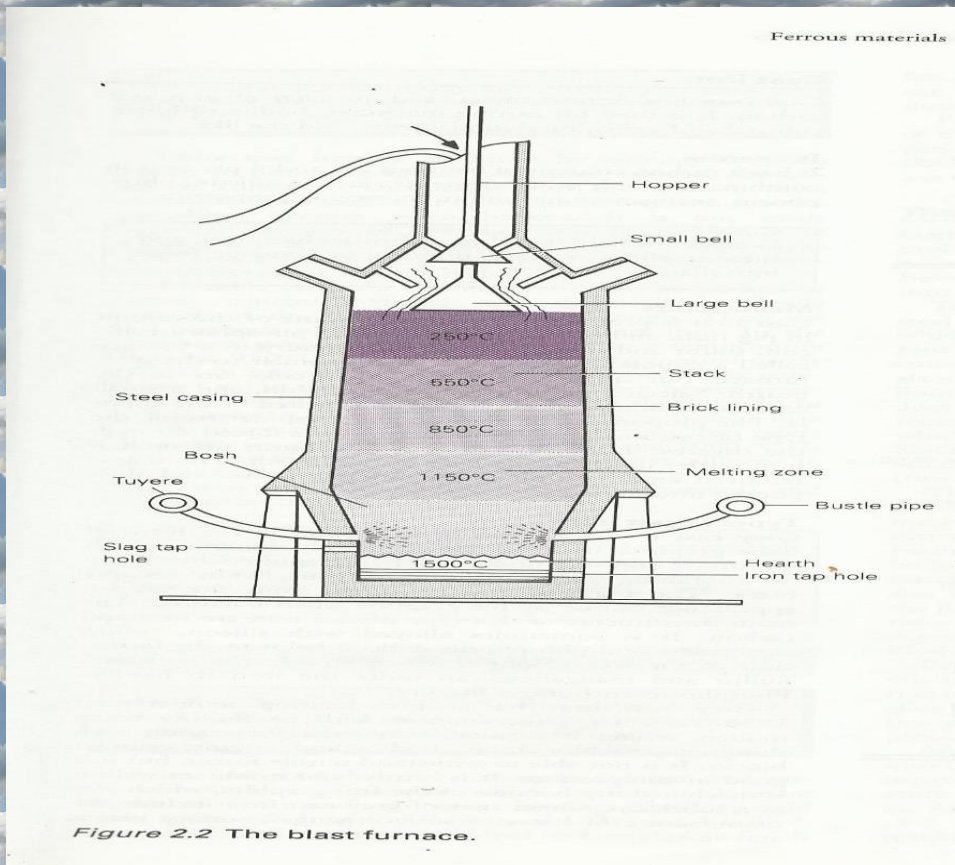
Production of iron

-iron is the most commonly used metal in everyday manufacture. This is because of its availability and low cost.

- It has great strength, easily cast into various shapes and has suitable alloying properties

Iron bearing rocks are called iron ores.

Ores are mixtures of minerals and stones dug from the ground. The blast furnace is used for separating iron from the ore by a process called smelting



- The blast furnace is a large steel shell lined with heat-resistant bricks.
- The materials which are called charge and which are needed for smelting are :
 - iron ore (raw material)
 - Coke(fuel)
 - Limestone (fluxing agent or cleaner)
- The charge (iron ore, coke and limestone) is put into the furnace through double bell lids opened one at a time to prevent furnace gases escaping into the atmosphere.
- The hot air (550c) from the hot blast stoves is blown into the bustle pipes and the tuyeres into the hearth of the furnace. This air causes the coke to burn fiercely and provides the necessary heat. As heating continues, iron separates and settles down the furnace in liquid form.
- At the same time, limestone combines with other impurities to form a liquid slag which floats on top of the molten iron.

- The slag is drawn through the slag notch, while the molten iron is drawn through the tapping hole.

Pig iron

When the molten iron from the blast furnace is put into moulds (containers) it forms blocks of iron called pigs and referred to as pig iron (ingots)

Production of steel

- the steel making process consists of removing excess carbon and other impurities e.g. silicon, manganese.
- sulphur and phosphorus from the pig iron and at the same time controlling the quantity of the elements other than iron that are an essential part of steel.

Steel is produced using one of the following methods

- Open hearth furnace
- The Bessemer converter furnace
- Basic oxygen furnace.
- Electric furnace

Open hearth furnace

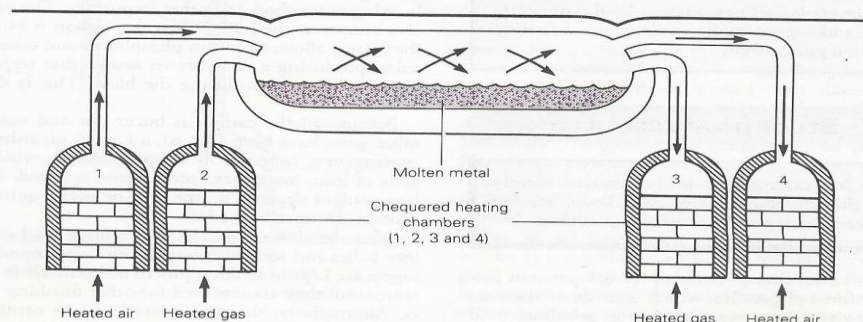


Figure 2.6 Open hearth furnace.

- The metal is contained in a shallow bath and in the front of the wall are the charging doors through which the raw materials are charged.
- The charge is made of molten pig, iron, limestone plus scrap metal. An air and gas mixture is heated by a very hot flame about 1650c melting the iron and burning out the impurities.
- when the required carbon content is reached, the molten metal is poured through the tapping hole into large containers forming what is known as ingot moulds.

- This steel is then allowed to solidify before being taken to rolling mills to be formed into various shapes and sizes.

The Bessemer converter

- The Bessemer converter is charged with molten metal and limestone.
- A blast of air is blown through the holes at the bottom oxygen burns out the impurities in the hot metal (iron).

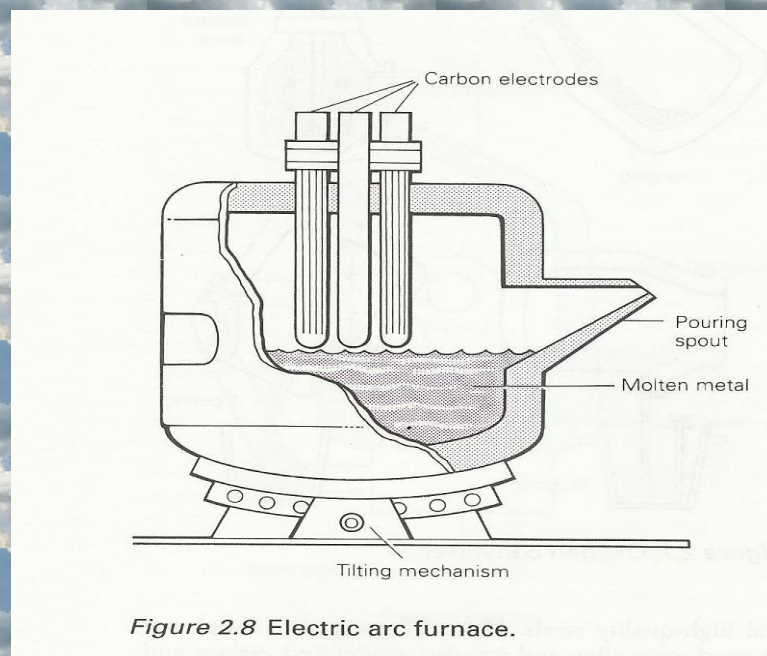
After 10 to 15 minutes the converter is tilted to pour the steel into a ladle (container)

- Then the alloying materials e.g. carbon, manganese and silicon are added to the steel to bring it to the correct chemical compound.

The basic oxygen furnace

- The basic oxygen furnace is charged with molten iron and scrap metal.
- The oxygen gas is used to burn out the impurities in the molten metal.
- An oxygen lance is lowered into the furnace and through it oxygen is blown at high pressure, onto the surface of the molten metal.
- This effects a very rapid conversion and produces good quality steel.
- The liquid steel is poured into a ladle through the pouring hole.

The electric furnace

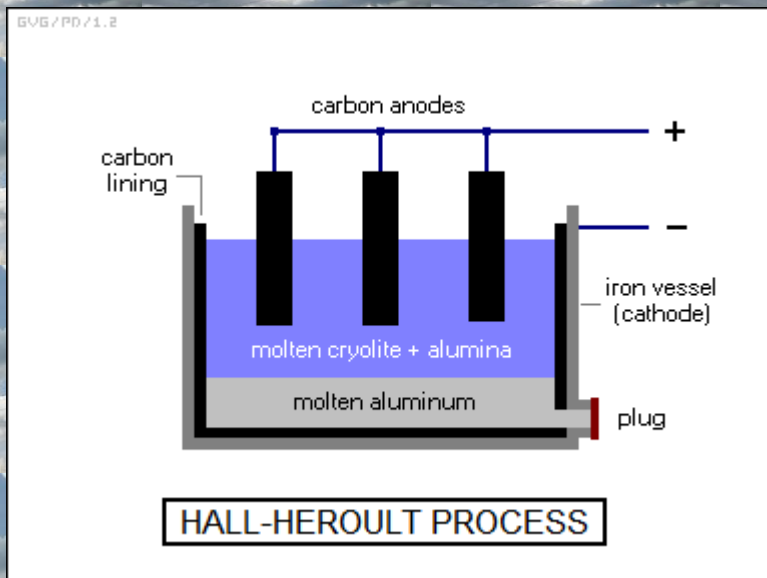


- The electric furnace is used to make high quality steel. This is because in it the heat and the amount of oxygen can be resulted. The three carbon electrodes are lowered until an arc strikes the metal.
- The heat generated by the electric arcs (3400) causes the metal to melt.

- Alloying materials such as chromium, nickel, tungsten etc are then added to make the type of alloy steel required.

Aluminum

- Aluminum ore is more abundant through the earth's crusts. All clays consist largely of Aluminum silicate but no satisfactory method has been developed whereby this and other aluminum compounds may be reduced to the metal in an economic manner.
- The only commercial source of AL is the bauxite (impure alumina or aluminum oxide)
- The alumina is then reduced to the metallic state
- Hall- herault process



- In this the alumina is stirred into a bath of molten cryolite (a fluoride of AL. and sodium) contained in a reduction cell.
- An electric current of some 30000 amperes at 6 volts flows between the anodes and the cathode.
- The heat from this maintains the cryolite in a molten state at 1000c.
- The molten cryolite dissolves the alumina and the passage of the electric current then decomposes it into AL. and O₂.
- The O₂ combines with the carbon of the anode forming co₂ which passes out of the top of the cell whilst the metal sinks to the bottom and is periodically drawn off and cast into ingots.
- As cast from the cell the metal is over 99% pure and a large proportion of it is used in this state
- This pure metal has little strength; small quantities of other metals are alloyed with it to improve its qualities.

Copper

- there are number of copper ores: malachite, containing about 55% copper
- Copper pyrites containing 35% copper, are the most commonly used ones.
- To extract the metal the ore is first crushed and sieved. It is then roasted to remove the sulphur.
- The claimed ore is then smelted in a reverberatory furnace to produce a malt of metal containing about 70% crude copper, the remainder being oxides and other impurities. The malt is poured into a converter and air is blown through the molten metal to oxidize the impurities.

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- The refined metal is then cast into slabs which are termed blister copper because of the blistered appearance of the melt.
- Blister copper is then further refined either by a second roasting in a reverberatory furnace to remove the combined oxygen or by an electrolytic process using anodes of blister copper a cathode of refined pure copper and concentrated copper sulphate for an electrolyte.
- The flow of electric current through the cell builds up pure copper on the anode, the impurities falling to the bottom as sludge.
- The product from this latter process is known as electrolytic copper
- **Bronzes**
- Tin is alloyed with copper to form a range of alloys called bronzes.
- Wrought tin bronzes contain up to about 8% tin, and can be cold-rolled or drawn. They can be softened by annealing and given a range of properties, depending upon the extent of cold-rolling after annealing.
- About 0.3 % phosphorus is included in wrought tin bronzes to produce wrought phosphor bronze, this type of bronze is used for springs, springy electrical contacts and for the suspension coils of moving – coil instruments.
- Casting tin bronzes contain between 10% and 18% tin and other elements.
- Casting phosphor bronze contains 0.05% phosphorus.
- It is a very good bearing metal because it combines hardness and toughness
- Small amounts of lead are included to improve the machinability of bronzes; about 5% lead also improves the bearing properties of bronzes.

Plastics

Plastics are made from petroleum (crude oil) coal and gases.

Advantages

1. They are light weight
2. They come in wide range of colors
3. They are available in different forms; as granules, as resin, in sheet forms etc.
4. They have good chemical resistance.
5. They form easily by cutting, casting, moulding etc.
6. They come in different textures.
7. Articles made from plastics need no surface protection.

Disadvantages.

1. They can be a fire risk.
2. Some types become brittle when exposed to the weather
3. They are expensive.

4. Some have low strength.

-Plastics are broadly grouped under two headings: thermoplastics and thermosetting.

Thermoplastic materials

-thermoplastic materials are the largest grouping.

They are hard and rigid but soften once heat is applied and the required temperature is reached.

-they can be easily molded.

-the common kinds of TM are as follows:

i) Cellulose nitrate

Comes in sheet forms tubes and rods.

It is tough, hard and burns with a camphor smell.

-Articles made from this type of material include table tennis balls, knife handles, drawing instruments such as set-squares etc.

ii) Cellulose acetate – is not as strong as celluloid.

-it is shaped easily by heating upon a strip heater or in an oven.

- It is used in making transparent wrappings, dolls, containers, welding goggles, machine guards.

- It comes in sheet, tube and rod forms.

iii) Polyvinyl chloride (PVC)

- Is tough, strong and resists abrasion.
- It's a good insulator.
- It's non flammable and non toxic.
- It can be obtained in flexible or rigid forms in sheets, tubes and extruded sections.
- It is used as insulation for electrical cables, baths, protective clothing TV parts and piping.

iv) Polythene (polyethylene (P.E)

Is tough, flexible and resists acids & alkalis. It is water proof. It is used for making washing bowls, electrical insulation, pipes, cups, dustbins and safety helmets.

iv) polytetrafluoroethane (PTFE)

- Is tough & flexible but weak.
- Use as non-stick coatings for cooking pans , used for making gaskets , seals , packing ,tank linings for chemicals *and* insulators.

vi) Nylon – *has good toughness and durability.*

-it is used for gears, zips and hinges because it has low frictional resistance and does not require lubrication. Can be used at temperatures between 50 and 150 °C

vii) Acrylic – is a clear rigid plastic which is lighter than glass.

It is a good insulator and is unaffected by dilute acids and concentrated alkalis.

It is used in place of glass for such applications as wind screens, machine guards, and roof lights & plastics lenses.

viii) ABS/ acrylonitrile butadiene styrene

Is very strong and tough and resists acids & alkalis.

- used for making tool handles, pipes and suitcases.

Thermosetting plastics

- T.P undergoes a chemical change during the process of being shaped.
- They cannot be reshaped by further heat and pressure polyester resin is an example.
- i) Phenol formaldehyde (bakelite) – it was the first synthetic plastic invented by L.H. Baekeland in 1909.
- It is a strong & rigid material which resists heat and serves as a good insulator.
- It is used for switches, electrical plugs, and pan handles.
- ii) Urea formaldehyde – is hard, rigid and absorbs water. It burns with difficulty and resists petrol, oil and grease.

It may be obtained in the form of powder or solution for laminating.

It is used for plugs, sockets, switches, electric iron handles etc.

iii) Polyester resin.

-Is strong, hard & tough and it resists dilute acid and alkalis.

It is usually reinforced with glass fibers and used in boat and car body production

Finishes and decorative process

- these processes are used to provide a good surface finish to the metal surface of the product
- In this either a negligible amount of metal is removed or a small amount of material is added to the surface of the product.

i) polishing

It is done with a very fine abrasive in loose form smeared on the polishing wheel with the work rubbing against the flexible wheel. A very small amount of material is removed in polishing.

ii) Electroplating

- An electrical process for depositing metal on a conductive surface that uses a cathode in an electrolytic bath containing dissolved salts of the metal being deposited is known as electroplating.

iii) cleaning .

Sometimes the blank surfaces need proper cleaning before being processed. For cleaning, pickling process is used. In this process sheets are immersed in a hot bath of 1 part dilute sulphuric acid and 20 parts water. After allowing sufficient time for pickling, the blanks are thoroughly washed in a stream of water and then allowed to dry.

iv) Enameling

-cooking utensils, service plates, machinery and fittings are often enameled.

-Enameling enables articles to withstand acids and oils. The process involves coating the surface of the work piece using special paints and baking it in an oven.

The finished article assumes a hard, glossy surface.

v) Lacquering

- Varnish or lacquer is applied to an already polished metal piece to protect the finish. Lacquer is applied by brushing, spraying or dipping. The process includes cleaning the item, warming it in an oven to decrease the viscosity of the lacquer or varnish and allowing it to cool.

vi) Etching

- This process involves the use of chemical solutions to change the surface aspect of the material.
- Etching is done without involving electric current and results in a dull, textured surface.

Advantages

Numerous patterns or designs can be achieved by masking off certain areas of a sheet with relatively high reflectivity and etching the exposed areas to reduce reflectivity.

Electrical materials and applications

Conductors: divided into two

i) Good conductors – such as silver, copper, Aluminum, which offer very little obstruction to the passage of current.

-they are good conductors of current and used for transmitting voltage and current within the generation and distribution of electricity.

ii) Poor conductors or resistors

-offer quite a considerable obstruction to the passage of current. The actual amount depending on the particular substance.

- e.g. Alloys such as manganin, eureka, constantin, german, silver, nomag and nichrome offer much resistance than iron and used to control the current in an electric circuit.

Certain rare metallic such as tantalum, osmium and tungsten offer extremely high obstruction. Used as filament in electric-light bulbs, being contained in a bulb exhausted of air so as to prevent them oxidizing & burning away.

Insulators

- It offers obstruction to the passage of current
- Examples glass, porcelain, rubber, shellac, mica, oiled silk, empire cloth, oils, resins, bitumen, paper etc.
- PVC materials (polyvinyl chloride), csp (chloro-sulphonated polyethylene are used for cable insulation.
- At temperatures near freezing pvc becomes stiff and more brittle and is liable to crack while at high temperature it becomes soft

Semiconductors

Are materials which are neither conductors nor insulators?

-silicon & germanium are semi-conductors and their resistivity depends upon their purity.

If a crystal of silicon has a very small amount of antimony or indium added as an impurity its resistivity is lowered and the greater the amount of impurity the lower the resistivity.

- Used in constructing semiconductor devices e.g. diode.

Cables

A cable to conduct an electric current consists of an inner core of copper or aluminum covered with an insulating sheath.

-Welding cables have to carry quite large currents and must be very flexible and as light as possible and to give this flexibility the conductor has very many strands of small diameter.

Metal forming processes

Forging – defined as the controlled plastic deformation of metal at elevated temperatures into a predetermined size or shape by using compressive forces.

Compressive forces are exerted through some type of die by a hammer, a press or by forging machines.

-forging processes are used in the manufacturing of the parts of tractors, automobiles, shipbuilding, cycle industries, rail road components, agricultural machinery etc.

Advantages of forging.

1. The forging have high ductility and offers great resistance to impact and fatigue loads.

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2. It refines the structure of the metal.
3. Forging distorts the previously created unidirectional fiber (created by rolling) and increases the strength by setting the direction of grains.
4. It results in considerable saving in time, labor and material as compared to the production of similar item by cutting from a solid stock and then shaping it.
5. The reasonable degree of accuracy may be obtained.
6. The forging may be welded.

Disadvantages

1. The initial cost of dies is high.
2. The maintenance cost is also high.
3. The rapid oxidation of metal surface at high temperature results in scaling which wears the dies.
4. The close tolerances are difficult to maintain.
5. Some materials are not readily worked by forging.

Forgeable materials in order of increasing forging difficulty

-aluminum alloys

- Magnesium alloys.

-copper alloys

-carbon and low alloys steel

-stainless steel

-nickel alloys

Titanium alloys

molybdenum alloys

tungsten alloys

beryllium

Bending

Is the plastic deformation of metals about a linear axis with little or no change in the surface area?

-the bars, rods, wires, tubes, and sheet metal may be bent to many shapes in the cold condition with the help of dies.

Bending dies

1. V-die

V dies are used in angle bending. Metal sheet is placed between punch and die and then punch is pressed against the die and transferring the angle to the sheet.

2. Wiping die – Are used for simple 90c bends

3. U die – is used in bending the sheet in u shape

4. Air bending die

5. Press brake dies.

6.4 – slide machine dies.

7. Draw bending die

8. Compression bending die.

Folding

Folding bar

- come in lengths of 250 – 300mm and are used for bending sheet metal at right angles

Filing

-it is the use of a file to finish a flat surface.

-The following are basic principles of cross- filing

1. Hold the work firmly in the vice.

2. Allow a small part of the work to project from the vice and avoid vibration.

3. The work should normally be level with your elbow.

4. Apply pressure in the forward stroke only.

5. Observe the correct methods for handling files for heavy and light filing.

Casting

It is a process in which diverse metal articles are manufactured by means of melting a metal, pouring it into prepared cavities called moulds and allowing it to solidify to take the shape or the form of the mould.

-There are two types of casting techniques:

i) Sand casting

ii) die- casting

Sand casting

- The process of sand casting takes three stages.

- Prepare ring the pattern, preparing the mould, melting and pouring the molten metal into the mould.
- A pattern is the exact shape or replica of an article to be cast.
- A mould is the cavity or impression of a pattern left in a box of sand into which a molten metal is poured.

Die- casting

It is a process for producing castings by injecting or forcing molten metal into a permanent metallic mould called a die.

Threading

The term for cutting internal threads by hand is tapping. The procedure is as follows.

1. Drill a hole that has about the same diameter as the core of the tap.
2. Hold the work firmly in the vice.
3. Grip the first tap in the tap wrench or holder.
4. Insert the tap squarely in the hole and turn it clockwise to start threading.
5. Turn anti-clockwise about a quarter – turn to break the swarf.
6. Apply cutting oil through the flutes, to ease cutting and wash away the chippings.
7. Repeat operation 5 until the entire length of thread on the tap has gone through (if it is through hole)
8. Follow the same procedure using the second and plug taps

The procedure for cutting external threads is as follows:

1. Square the end of the work and chamfer it (using a file grinding machine or centre lathe for an easy start.
2. Grip the die, held in the stock, firmly and squarely on the work.
3. turn clockwise, about a quarter-turn and ease back to remove chippings.
4. Apply a good supply of oil (lubricant)
5. Make adjustments of the screws after making a full cut until the depth required is achieved.

METAL- SHOP TOOLS AND MEASUREMENTS

1. Scales

- A scale drawing is a drawing in which the outline form and various details have been drawn to exact shape and proportion with the use of drawing instruments

- In any scale drawing the measurements are made with a scale or ruler.

The scales used for metric drawing are;

- (a) full size
- (b) reduction scale
- (c) enlargement scale

Linear scale – involve a single dimension

Non linear scale – involves more than one dimension.

2. Tolerance – the difference between the limits (dimensions) of a shaft or hole

3. Limits – are the two extreme sizes for a given dimension.

They represent the largest permissible size for a hole or shaft and the smallest

4. Fits – refer to a system of mating shafts and holes.

The 3 basic types of fit in this system are clearance, transition and interference. With a clearance fit, the size of the hole is greater than that of the shaft.

- With transition fit, the sizes of the hole and the shaft are the same. A light force required for assembly
- With interference fit, the lower limit of the shaft greater than the upper limit of the hole.
- Considerable force is required for assembly.

Marking out techniques

Line and measurement

- Length is the distance between two lines and termed as line standards.
- Length is also define as the distance between two parallel plane surfaces and this is called end standard.
- When lengths, depth, heights and diameters are to be measured it is more convenient to use equipment that can be used for intermediate sizes, without the need to build up.

The engineers rule

- The sizes are indicated by fine lines scribed on a steel strip and so this is a line measurement system.
- The steel rule is made of spring steel.
- The common lengths are 150mm and 300mm.
- Metric rules are graduated in millimeters (mm) and centimeters (cm).

Fig (a). The steel rule

- when using the rule , take measurements from datum side or surface as shown in fig (c)

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Pocket or flexible rules

-have blades made of steel which are spring-loaded into a case. They are longer than steel rules and can therefore be used to measure much greater lengths (typically up to 3m)

They are portable and can be carried about in the pocket.

- It is difficult to take accurate readings with a rule because you have to estimate measurements.
- Better accuracy can be obtained by a vernier caliper

Vernier caliper

This consists principally of

1. A rule scale (the fixed scale)
2. A fixed jaw (part of the rule scale)
3. A vernier scale (a moving scale)
4. A sliding jaw (attached to the moving scale)

The vernier caliper

- the rule scale is graduated in mm
- the moving scale moves on the rule scales
- attached to the sliding jaw is clamp
- The vernier scale is graduated to read up to 49mm. there is 50 divisions which mean that there is difference of 0.02mm between the vernier scale and the main scale.
- Taking a measurement with the vernier caliper
 1. Moving the sliding jaw to be in contact with the faces being measured.
 2. tighten the locking screw on the clamp
 3. Make fine adjustment using the fine – setting screw.
 4. Move the jaws so that they just touch the work do not apply any force.
 5. Tighten the head lock.

Main scale reading = 25.00mm

Vernier scale reading = 22 x 0.02 (0.44mm)

Final reading = 25.44mm

- Vernier caliper is a useful tool for taking external and internal measurements.
- Add the widths of the jaws (stated on the caliper) when taking internal measurements.
- A is a rule of any length graduated in 1mm and 0.5mm divisions. B is a short sliding scale with twenty-five (25) equal divisions which equal twenty four of the 0.5mm divisions on scale A.
- the difference in length between a division on A and one on B

Will therefore be $12/24 - 12/25 = 0.5 - 0.48 = 0.02 \text{ mm}$

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A micrometer

- The micrometer has a thread with pitch 0.5mm.

This means that the spindle advances by 0.5mm

For each turn. However there 50 graduations on the thimble .so the movement advanced for each graduation of the thimble is $0.5/50 = 0.01\text{mm}$

- the metric micrometer is able to measure ranges of 25mm (i.e. for 0-25 mm, 25-50mm etc)

The procedure for using the micrometer

1. Hold the plastic insert to prevent thermal expansion
2. Keep the measuring faces square with the surfaces that you are measuring to ensure an accurate measurement.
3. Turn the thimble until the faces touch the work.
4. Use the ratchet to obtain the correct pressure when turning the thimble and prevent the jaw moving further when it comes into contact with the work.

Micrometer reading

Handling of the micrometer

1. Store the micrometer away from dust
2. Clean the measuring faces for good results.
3. Oil the micrometer regularly to avoid rust.
4. Pack the micrometer in its box when not in use
5. Do not use force on the thimble or ratchet.